A WEB-BASED BANANA LEAF HEALTH DETECTION AND FEEDBACK LEARNING SYSTEM USING A CONVOLUTIONAL NEURAL NETWORK

A PROJECT PROPOSAL FOR CS 20/L
CS PROFESSIONAL TRACK 5

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BACKGROUND OF THE STUDY

Banana production is a major agricultural activity in the Davao Region. The Philippine Statistics Authority reported that banana accounted for 94.8 percent of Davao Region fruit crop production in 2023, equivalent to 3,277,125.80 metric tons. In 2024, fruit crop production in the region declined by 2.2 percent, and the decrease was largely attributed to reduced banana output. These figures show the local importance of banana crop health and justify the need for practical monitoring tools in the Davao and Tagum context.

Banana leaves can show visible symptoms caused by disease, nutrient stress, environmental stress, and physical damage. Field inspection remains important, but it can be time-consuming and dependent on the observer's experience, especially when symptoms are subtle or when many plants must be checked. Recent studies on banana plant health and plant disease detection show that computer vision and deep learning can support crop monitoring by analyzing image characteristics such as color, texture, edge structure, and overall leaf appearance.

This proposal focuses on improving the current Banana Leaf Detector, a web-based application that accepts an uploaded image and classifies it as Healthy Leaf, Unhealthy Leaf, or Non-Leaf. The existing prototype uses K-Nearest Neighbors (KNN) with manually extracted color, texture, and shape features. This study proposes to upgrade the classifier to a Convolutional Neural Network (CNN) using transfer learning, specifically a frozen ResNet18 backbone with a trainable classification head. In addition to prediction, the prototype will retain feedback logging, scan history, and analytics features to support future dataset improvement and project continuation.

The research gap addressed by this study is the need for a low-resource and understandable prototype for banana leaf health screening. Existing deep learning studies often focus on high-performance models or broad agricultural datasets, while this project aims to improve an existing KNN-based prototype into a practical CNN-based system that can be documented, tested, and presented within the CS20/L course requirements.

OBJECTIVE OF THE STUDY

The main objective of this study is to improve the current KNN-based Banana Leaf Detector by developing and evaluating a CNN-based image classification approach for screening uploaded images and supporting feedback-based dataset improvement.

- To develop a browser-based interface that accepts valid image uploads and returns banana leaf health screening results.
- To classify uploaded images into Healthy Leaf, Unhealthy Leaf, or Non-Leaf categories.
- To preprocess uploaded images through RGB conversion, resizing to 224 x 224 pixels, tensor conversion, and normalization suitable for CNN inference.
- To train and evaluate a CNN classifier using transfer learning with a frozen ResNet18 feature extractor and a trainable classification head.
- To display prediction confidence and a brief explanation of the model output.

- To log user feedback and corrected labels through Firebase Firestore for possible dataset improvement.
- To summarize scan history, classification activity, and feedback results through basic analytics.
- To evaluate the prototype using accuracy, precision, recall, F1-score, confusion matrix, and error analysis.

METHODOLOGY

This section presents the methods employed in conducting the study. It covers the research design, target business entity, research instruments, procedure, and data analysis techniques.

Research Design - This study uses an applied developmental and quantitative experimental research design. The developmental component covers the improvement of the existing Banana Leaf Detector web system from a KNN-based classifier to a CNN-based classifier, while the quantitative component covers the evaluation of the CNN model using labeled image data and standard classification metrics.

Target Business Entity - This study does not use a single private company as the data owner. The intended application context is local banana growers, agriculture students, information technology students, and agricultural extension personnel in Davao Region or Tagum City who may benefit from a simple banana leaf screening tool.

Research Instruments - The primary research instrument is the Banana Leaf Detector web system. The current system uses Python, Flask, OpenCV, NumPy, Pandas, scikit-image, scikit-learn, HTML, CSS, JavaScript, Firebase Firestore, and a labeled banana leaf dataset. The proposed improvement will add PyTorch and torchvision for CNN training, testing, and inference using the available image dataset.

The study will be conducted through the following procedure:

1. Define the project scope and target classes: Healthy Leaf, Unhealthy Leaf, and Non-Leaf.
2. Prepare and review the labeled image dataset to ensure label consistency, image readability, duplicate reduction, and reasonable class distribution.
3. Implement or verify image preprocessing, including file validation, secure upload handling, RGB conversion, resizing to 224 x 224 pixels, tensor conversion, and ImageNet-style normalization for transfer learning.
4. Train the CNN classifier using the ResNet18-based transfer learning architecture and evaluate its validation and test performance.
5. Integrate the trained CNN model into the Flask web application and test the upload, prediction, explanation, feedback, history, and analytics functions.
6. Configure Firebase Firestore for storing feedback records and related history data.
7. Document the system architecture, workflow, evaluation results, limitations, and future improvements.

Data Analysis Techniques - The study will use supervised image classification analysis. Model performance will be interpreted using accuracy, precision, recall, F1-score, confusion matrix, and qualitative error analysis. Training and validation behavior will also be reviewed to assess the model's consistency and limitations.

Data Handling and Ethical Considerations - Only plant images, prediction outputs, and feedback labels are necessary for the proposed CNN-based study. Existing extracted numerical features may be used for comparison with the previous KNN baseline, but CNN training will primarily use labeled image files. The system should avoid collecting unrelated personal information. If images are gathered from farms or private locations, permission should be obtained before use. File upload validation and secure filename handling will also be applied to reduce upload-related risk.

EXPECTED RESULTS AND DISCUSSION

The proposed study is expected to demonstrate that the Banana Leaf Detector can be improved from a KNN-based feature classifier into a CNN-based web prototype. The improved system should accept an uploaded image, preprocess it for CNN inference, classify it as Healthy Leaf, Unhealthy Leaf, or Non-Leaf, and present the result with confidence scores and a short explanation. This expected output will show that the project is not only a model training activity, but also a usable web application that connects image upload, preprocessing, prediction, feedback, history, and analytics features in one workflow.

The proposed CNN implementation is expected to show that transfer learning is a practical approach for this project. By using a frozen ResNet18 feature extractor and a trainable classification head, the system can learn visual patterns from the project dataset without requiring the team to train a large deep learning model from the beginning. The expected model checkpoint, training curves, classification report, and confusion matrix will provide evidence that the CNN training workflow was completed and prepared for integration into the Flask application. This supports the research objective of developing an improved banana leaf health detection prototype using CNN image classification.

The expected findings are aligned with recent literature showing that CNN-based and deep learning approaches are useful for plant disease and crop health detection. Related studies report that convolutional models can identify visual disease patterns when enough representative image data are available. However, the proposed project also recognizes the limitations discussed in the literature: model performance depends on image quality, dataset size, class balance, lighting conditions, background noise, and symptom variation. Therefore, the prototype should be interpreted as a preliminary screening and learning tool rather than a replacement for expert agricultural diagnosis.

Overall, the proposed project is expected to demonstrate that a CNN-based improvement can support banana leaf health screening and feedback collection within the CS20/L project scope. The system will meet the study objective if it can run locally or on a supported deployment platform, accept valid image uploads, classify images through the CNN model, display understandable prediction output, and store feedback and history records. Future work should focus on collecting more diverse banana leaf images, validating the model with a larger test set, improving field-image robustness, and comparing the CNN results with other lightweight models or mobile-friendly architectures.

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